PdfHelperReader – Comprehensive User Manual

(v4.2.0) • High-Performance Dual-Pane Reading, Refined Vocabulary Engine & Integrated Research Environment

1. Welcome & Introduction

PdfHelperReader is a specialized Android document reading environment designed for language learning, academic research, and deep study. It reads both PDF and EPUB documents, pairing your main book with a fine-tuned Refined Vocabulary sidecar PDF or an embedded Google Web Search browser in real-time.

2. Upper Main Ribbon Controls (Top Action Bar)

Button	Function & Detailed How-To-Use Guide
B	**Add Bookmark**: Tap to bookmark your current reading position. Captures creation timestamp and extracts the first 25 words from the book page.
BL	**Bookmark List**: Opens saved bookmarks dialog. Shows creation date/time and 25-word book page snippet previews for the current book.
HL	**Highlights List**: Opens interactive highlights dialog. Displays unified multi-page highlights (Pages X-Y) and single-page highlights with full joined excerpt, jump-to-page navigation, and single-tap deletion.
Both / Book / Vocab	**Split Mode Toggle**: Cycles screen layout between Both (dual-pane split view), Book (full-screen main document), or Vocab (full-screen vocabulary pane).

Button	Function & Detailed How-To-Use Guide
Z	**Zen Mode & Gestures**: Hides toolbars and system bars for 100% edge-to-edge reading. **Triple-tap** anywhere in normal mode to enter Zen Mode with hidden buttons; **triple-tap** again to return to normal mode. In Zen Mode, **double-tap** to hide/unhide the right-edge sidebar controls.
SR	**Synced Reading**: Toggles 1:1 real-time bidirectional scroll synchronization. When scrolling either the Book or Vocab pane, the other pane scrolls automatically in lockstep.
AS	**Auto Scroll**: Gentle continuous frame-by-frame auto-scroll calibrated at 1 viewport height / 5 minutes. Touch dragging is supported without stopping auto-scroll.
Sync	**Page Sync**: Instantly jumps the Vocabulary pane to match your current Book page using the exact vocab_map array.
G:Vocab / G:Google	**Vocab Mode & Preserved Search State**: Toggles lower pane between Vocabulary PDF sidecar and embedded Google Search. Automatically switches to Dual-Pane Split View with persistent layout and pre-warmed Chromium engine. Preserves browsing history and state when switching back and forth, and performs new searches on-demand.
T	**Theme Toggle**: Switches Dark/Light mode with high-contrast ColorMatrix PDF page color inversion for comfortable night reading.

Button	Function & Detailed How-To-Use Guide
R:C / R:P	**Reading Mode**: Toggles Continuous vertical scrolling (R:C) or fit-to-width Page-by-Page mode (R:P).
V:PDF / V:TXT	**View Mode**: Toggles vector PDF page rendering with transparent OCR text selection overlay (V:PDF) or plain text extraction (V:TXT).
H	**History (Dual-Tab)**: Opens history dialog with two tabs: **Reading History** (tracks First Opened & Last Closed timestamps for up to 1,000 documents with Del vocab) and **Search History** (logs all past Google searches with one-tap re-search, item deletion, and Clear All). Tab selection and scroll positions are permanently preserved.
O	**OCR Generator**: Applies Google ML Kit OCR to image-based or scanned PDFs, adding a searchable text layer saved to Downloads/PdfHelperReader_files/, with Force OCR mode to override existing text.
LOB	**Load Online Book**: Opens dual-tab online library (35 GitHub CDN classics + official Project Gutenberg search engine for 70,000+ books with EPUB and PDF download options).
OB / OV / LV	**Open Book / Open Vocab / Load Vocab**: Open PDF or EPUB book/vocab files via system file picker, or generate vocabulary PDF automatically.
Update	**In-App Update**: Checks GitHub latest release API with 3-tier fallback, streams live download progress, and launches Android Package Installer seamlessly.

Button	Function & Detailed How-To-Use Guide
App Info	**Developer Info**: Displays developer profile (Ohid Sajol), direct clickable GitHub repository link (github.com/ohidsajol/PdfHelperReader), and full application overview.

3. Refined Vocabulary Engine & Difficulty Scores

PdfHelperReader features a fine-tuned Refined Vocabulary database containing over **964,000 entries**.

• **1.5× Enlarged Typography**: Generated vocabulary PDFs render with 1.5× enlarged typography (24pt headers, 19.5pt headwords & scores, and 16.5pt definitions, examples, synonyms, and antonyms) with generous proportional line and block spacing for effortless study on mobile screens.

• **Headword References**: Companion vocabulary PDFs explicitly display cross-references and inflected headword targets (e.g. **wobble (Ref: Wabble) (32.5)**) in high-contrast italic slate typography.

• **Difficulty Scores**: Word rarity scores (1–100) are displayed right beside headwords in slate gray (e.g. **anopheles (29.25)**).

• **Word Frequency Filtering**: An interactive slider dialog lets you filter vocabulary density (1 to 100) before generating the sidecar PDF.

• **Instant Definition & Synonyms Popup**: Selecting any recognized word in the reader opens a lower-pane quick popup displaying part of speech, definition, synonyms, and antonyms. The popup always renders in the lower pane during split view (even when Google is open) and auto-dismisses smoothly when opening Google search.

• **Free Online Dictionary Fallback**: If a selected word is missing from the local offline dictionary, the reader queries the free online dictionary API in the background, displaying definitions, synonyms, antonyms, and an **online** badge.

4. Immersive Fullscreen Zen Mode & Gestures (Double & Triple Tap)

Activating Zen Mode hides all toolbars as well as the system status bar and navigation bar for 100% immersive edge-to-edge reading.
• **Triple-Tap in Normal Mode**: Instantly enters Zen Mode with sidebar buttons hidden for a totally clean distraction-free view.
• **Triple-Tap in Zen Mode**: Instantly exits Zen Mode directly back to Normal Mode without showing the sidebar buttons.
• **Double-Tap in Zen Mode**: Toggles (hides/unhides) the right-edge sidebar controls on demand.
Control strip visibility state is preserved across screen orientation changes.

5. Unified Multi-Page Highlights & Selection Management

When selecting and highlighting text that spans across multiple consecutive pages (e.g. Pages 31-34), PdfHelperReader automatically unifies them into a single coherent entry in the Highlights dialog (**HL**). The entry displays the full joined text, a multi-page range badge, jumps directly to the starting page when tapped, and deletes all linked page segments across the entire document in one action.

6. Smooth Center-Anchored Two-Finger Pinch-to-Zoom

Fluid two-finger pinch-to-zoom allows seamless magnification on both single-page (**PageView**) and continuous vertical scroll (**ContinuousPageView**) modes. The zoom gesture anchors precisely around the center point of each pane, spreading outward symmetrically in all directions with a silky **0.085f** smoothing factor.

7. Instant Copy-to-Search Popup Banner

Whenever you select and copy text anywhere in the app or system, a floating popup banner appears allowing you to search the copied string in Google with a single tap. The banner responds reliably to every copy action, even when re-copying the same selection.

8. In-App Updates & Interactive Developer Info

Tapping **Update** checks GitHub for new release versions with a 3-tier fallback (API, Web Redirect, Feed), downloads the APK in the background with a live

progress dialog, and launches the Package Installer. Tapping **App Info** displays developer details and an interactive clickable link taking you directly to the GitHub repository.

9. Gutenberg Books Search Engine (LOB Tab 2)

Tapping **LOB** opens the dual-tab Online Library. The Gutenberg Books tab features live search across 70,000+ books with direct **[EPUB]** and **[PDF]** download options.

10. Full EPUB eBook Support

PdfHelperReader unzips and converts eBooks on-demand into clean, typeset vector documents with large 18pt typography. Converted files are cached permanently in *Downloads/PdfHelperReader_files/* so re-opening an EPUB loads instantly in 0ms.

11. 500 MB Storage Manager

An automatic 500 MB Storage Manager monitors folder size, deleting older files step-by-step while protecting the latest active file in each category.

12. Dual-Tab History & Persistent App State

The History dialog (**H**) features two independent tabs: **Reading History** and **Search History**.

• **Search History**: Tracks all past Google searches. Tapping any item re-executes the search instantly, with single-tap deletion and a **Clear All** option.

• **Tab State & Scroll Persistence**: Tab selection and exact scroll positions are preserved across dialog dismissals and app restarts.

• **Ribbon Toolbar Position Preservation**: The horizontal scroll offset of the upper ribbon toolbar is saved permanently and restored smoothly across Zen mode toggles and app restarts.

13. Precision Sub-Word Selection & Smart Search Formatting

• **Precision Drag Selection**: Long-pressing on any word expands the selection to cover the entire word, while dragging the selection handles allows

fine-grained, sub-word precision without rigid word snapping.
• **Smart Search Query Formatting**: Initiating a Google search from selected text automatically crafts the search query: single words search *meaning of this: "word"*, while multi-word phrases search *meaning of this part by part: "phrase"*.